TECHNICAL ROCKETRY REPORT

Parametric Evaluation of KNSB Solid Rocket Propellant

Subject: Part 1 — Potassium Nitrate Quality and Ballistic Impacts

Source Data: Richard Nakka's Experimental Rocketry Database

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1. Potassium Nitrate Quality and Ballistic Impacts on KNSB

The performance, structural integrity, and combustion predictability of Potassium Nitrate/ Sorbitol (**KNSB**) propellant are heavily dependent on the chemical purity and physical condition of its primary constituent, Potassium Nitrate (**KNO_3**), which accounts for 65% of the standard formulation by mass. Empirical research conducted by Richard Nakka highlights clear relationships between oxidizer contaminants and variations in target ballistic performance.

1.1 Purity Grades and the Practicality of Fertilizer-Grade Oxidizers

While technical or reagent-grade Potassium Nitrate (purity of 99.5% or higher) offers the highest baseline consistency, experimental testing demonstrates that industrial or agricultural fertilizer-grade Potassium Nitrate (typically sold as 14-0-45 or prilled chemical) is highly viable for amateur experimental rocketry.

Fertilizer-grade **KNO_3** generally exhibits a purity of 98% to 99%. Ballistic static test comparisons reveal that this 1% to 2% variance in purity does not drastically alter the specific impulse or total impulse of the motor, provided that the material is adequately processed to eliminate coarse structural binders or hygroscopic anti-caking agents.

1.2 Chemical Impurities and Diagnostic pH Verification

Low-grade or poorly manufactured **KNO_3** frequently contains underlying chemical contaminants, primarily sodium chloride (**NaCl**), sodium nitrate (**$NaNO_3$**), or residual alkaline compounds introduced during synthesis or anti-caking spraying. These additives can act unpredictably within the combustion zone, either serving as accidental secondary catalysts or local combustion suppressants. This leads to severe deviations from predicted chamber pressure profiles.

A reliable diagnostic metric developed by Nakka involves assessing the pH profile of a saturated aqueous solution of the oxidizer sample:

- **High-Quality / Pure Sample:** Displays a neutral to slightly acidic baseline profile (pH approximately 6.0 to 7.0).

- **Contaminated Sample:** Displays an alkaline profile (pH greater than 7.0). This high pH strongly indicates the presence of basic salt impurities or chemical surfactants that will disrupt consistent thermal processing and burn rate boundaries.

> **CRITICAL TECHNICAL INSIGHT: MOISTURE AS AN ACTIVE CONTAMINANT**
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> Moisture represents the most destructive and prevalent impurity in solid rocket candy propellants. Both KNO_3 and Sorbitol exhibit highly hygroscopic behaviors, actively absorbing moisture from the surrounding atmosphere when relative humidity is elevated. Residual or absorbed water acts as a potent internal thermal sink, profoundly altering both physical castability and ballistic parameters.

1.3 Ballistic and Physical Degradation from Moisture Seepage

Experimental trials reveal three catastrophic vectors through which residual moisture alters KNSB characteristics:

1. **Burn Rate Suppression:** Water absorbs substantial thermal energy via vapor phase transitions during combustion, suppressing the localized flame front. A moisture content as small as 1.0% by mass significantly decreases the linear regression rate of the propellant, leading to systemic under-pressurization or structural chuffing (oscillatory, unstable start-up combustion cycles).
2. **Thermodynamic Efficiency Loss:** The presence of 1.0% water content correlates directly to an approximate 1.0% loss in theoretical Specific Impulse due to the reduction of core combustion temperatures and a higher molecular weight of the exhaust gas products.
3. **Mechanical Failure Modes:** When KNSB slurry is cast with a trace water percentage, the cooled solid mass displays highly anomalous physical traits. The grain remains pliable and easily deformed under slow pressure, yet it becomes exceptionally brittle under high-velocity structural loading. If dropped or subjected to rapid ignition pressurization forces, the grain readily cracks, exponentially expanding the combustion surface area and resulting in catastrophic motor casing failure.

1.4 Technical Remediation: Purification via Recrystallization

To salvage low-grade or highly contaminated agricultural KNO_3, Richard Nakka prescribes a precise thermal recrystallization methodology. This mechanical process exploits the steep solubility differential of Potassium Nitrate in water across distinct temperature boundaries (highly soluble at boiling points, poorly soluble near freezing points):

> **Nakka Recrystallization Ratio Parameters:**
> • Target Crude Oxidizer: 1.0 kg
> • Solvent Medium (Distilled H_2O): 500 ml
> • Thermal Target: Dissolve entirely at a full boil of 110°C

The boiling solution is filtered through a fine sieve to isolate insoluble particulate contaminants, then allowed to cool slowly to room temperature. This slow cooling initiates an exothermic crystal growth phase, selectively binding **KNO_3** ions into a clean crystalline network while leaving soluble contaminants trapped within the supernatant liquid (mother liquor).

The solution is refrigerated overnight, mechanically compressed inside a fabric sheet to express the maximum liquor volume, washed cleanly with 500 ml of ice-cold distilled water to rinse surface salts, and air-dried at ambient temperature. *Oven drying must be strictly avoided*, as residual moisture will cause the chemical to re-dissolve into a dense, non-usable mass.

VERIFIED REFERENCE LITERATURE

This technical report synthesizes empirical findings, raw data coordinates, and safety methodologies derived directly from the laboratory records of Richard Nakka's Experimental Rocketry database:

- **Oxidizer Characteristics & Modifiers:** nakka-rocketry.net/oxidex.html
- **Oxidizer Recrystallization Protocols:** nakka-rocketry.net/knpurify.html
- **KNSB Formulation Frameworks:** nakka-rocketry.net/kn-exp.html